Cleveland Clinic

Acid Reflux & GERD

Acid reflux might happen after a large, rich meal, or when you lie down too soon after dinner. Stomach acid rises into your esophagus, causing heartburn and other symptoms. Occasional acid reflux is manageable at home, but chronic acid reflux (GERD) might need treatment. GERD can damage your esophagus tissues over time.

Overview

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Stomach acid may move backward into the esophagus when the lower esophageal sphincter is weak.

What is acid reflux?

Your stomach contents are supposed to travel only one way: down. When acid from inside your stomach flows backward — meaning, up — into your esophagus and throat, it's called acid reflux.

When acid creeps into places it doesn't belong, you're bound to feel it. Acid irritates and inflames the tissues inside your esophagus, which runs from your stomach up through your chest to your throat.

Almost everyone has experienced an occasional episode of acid reflux. It might feel like indigestion — burning stomach pain after eating — or heartburn — burning chest pain close to your sternum.

Occasional acid reflux is uncomfortable, but it's not a disease. But some people have reflux all the time. Chronic acid reflux can really affect your quality of life, and it can also do real damage to your tissues.

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What's the difference between GERD and stomach ulcers?

What is GERD?

GERD stands for gastroesophageal reflux disease. Depending on where you live, it may be spelled GORD for gastro-oesophageal reflux disease. However you spell it, GERD is chronic acid reflux in your esophagus. Acid reflux is considered chronic when you've had it at least twice a week for several weeks.

Temporary conditions can cause temporary acid reflux. But GERD is a constant, mechanical problem. It means that the mechanisms that are supposed to keep acid out of your esophagus aren't working right.

How common are acid reflux and GERD?

Occasional, uncomplicated gastroesophageal reflux (GER) and gastroesophageal reflux disease (GERD) are both common. GERD is estimated to affect about 20% of adults and 10% of children in the U.S.

Symptoms and Causes

What are the symptoms of acid reflux and GERD?

Symptoms of acid reflux and GERD may include:

- **Backwash**. You might notice acid, food or liquids backwashing from your stomach into your throat after eating. This is also called regurgitation. You might notice the sour taste of the acid.
- **A burning feeling**. Acid literally burns the tissues in your esophagus. If it feels like it's in your chest, it's called heartburn. If it feels closer to your stomach, you might call it acid indigestion.
- **Noncardiac chest pain**. Some people feel pain in their esophagus that doesn't feel like burning. Esophagus pain triggers the same nerves as heart-related pain does, so it might feel like that.
- **Nausea**. Acid overflow or backwash may make you feel queasy or make you lose your appetite. Although you may have eaten a while ago, it may feel like there's still more food to digest.
- **Sore throat**. If acid rises into your throat, it can make it sore. It might feel like there's a lump in your throat, or like it's hard to swallow. Reflux into your throat often happens at night.
- **Asthma symptoms**. GERD can trigger asthma-like symptoms, like chronic coughing, wheezing and shortness of breath. If acid particles get into your airways, it can make them contract.

GERD symptoms may be worse:

- At night or while lying down.
- After a large or fatty meal.
- After bending over.
- After smoking or drinking alcohol.

Do babies get GERD?

It's normal for babies to spit up (regurgitate). There might be some acid in it, but usually not enough to really bother them. GERD is a more severe condition that causes distress or complications with feeding.

Babies are more likely to have GERD if they're born prematurely or with a condition that affects their esophagus. A pediatrician might check for GERD if your baby seems to be struggling in some way.

Symptoms of GERD in babies (or small children) may include:

- Excessive fussiness.
- Difficulty sleeping.
- Refusing to feed.
- Small vomiting episodes.
- Wheezing or hoarseness.
- Bad breath.

What causes acid reflux?

For acid to get into your esophagus, it needs to get past the valve at the bottom of your esophagus that usually keeps things from coming back up. This valve is called your lower esophageal sphincter (LES).

Your LES is a circular muscle that opens when you swallow and then closes again to keep substances in your stomach. It also opens a little to let gas bubbles out when you're burping or have hiccups.

Acid reflux happens when your LES weakens or relaxes enough to let acid pass. Some temporary things can relax your LES, like lying down after a large meal. But if you have GERD, it means your LES is relaxing often.

Many things can contribute to weakening your LES, either temporarily or permanently. Sometimes occasional acid reflux turns into chronic GERD when these factors overlap or persist for a long time.

Common causes of acid reflux and GERD include:

- **Hiatal hernia**. A hiatal hernia happens when the top of your stomach pushes up through the hole in your diaphragm where your esophagus passes through. It squeezes in next to your esophagus, compressing them both and trapping acid. It also moves your LES above your diaphragm, where it loses some of its muscular support. Hiatal hernias are very common, especially as you get older. They usually occur gradually, and they can gradually worsen.
- **Pregnancy**. Pregnancy is a common cause of temporary acid reflux. The pressure and volume in your abdomen can push, stretch and weaken the muscles in your diaphragm that support your LES. Pregnancy hormones may also encourage your LES to relax. Pregnancy brings high levels of the hormone relaxin, which relaxes your muscles so they can stretch to make room for the fetus. It also brings high levels of estrogen and progesterone, which may also relax your LES.
- **Obesity**. Obesity increases the pressure and volume in your abdomen, which affects your LES similarly to how pregnancy does. Obesity also tends to last longer than pregnancy, which can weaken the muscles more permanently. It's a common contributing factor to developing a hiatal hernia. Since fat tissue secretes estrogen, having more of it also raises your estrogen levels.
- **Smoking**. Tobacco smoke relaxes your LES, whether you're the one smoking or you're exposed to second-hand smoke. Smoking also triggers coughing, which opens your LES. Smoking and chronic coughing can weaken your diaphragm muscles and contribute to developing a hiatal hernia. Smoking also slows down your digestion and causes your stomach to produce more acid.

Other possible causes of GERD include:

Birth defects. Congenital defects like esophageal atresia and hernias can affect your LES.

Connective tissue diseases. Diseases like scleroderma may affect your esophagus muscles.

Prior surgery. Surgery in your chest or upper abdomen may have injured your esophagus.

Medications. Certain medications can have a relaxing effect on your LES, including:

- Benzodiazepines, a type of sedative.
- Calcium channel blockers, which treat high blood pressure.
- Tricyclic antidepressants, which treat depression and pain.
- NSAIDs (nonsteroidal anti-inflammatory drugs) like aspirin and ibuprofen.
- Theophylline, a common asthma medication.
- Hormone therapy (HT) medications for menopause.

Can foods cause acid reflux?

Foods and drinks probably aren't enough to cause acid reflux alone, but they can contribute to it. Chocolate, coffee, alcohol, mint, garlic and onions may have a relaxing effect on your LES in higher doses.

Fatty foods increase stomach acid and take longer to digest, so there's more opportunity for acid to escape. If you have a heavier meal for dinner, it might not have time to digest before you lie down.

What are the possible complications of chronic acid reflux (GERD)?

Stomach acid is powerful stuff, built to break down the food you eat for digestion. Your stomach has a tough inner lining to protect it from its own acid. But your other organs don't have this protection.

Acid reflux mostly affects your esophagus, though sometimes it can get into your windpipe or even your airways. A little acid may just feel momentarily uncomfortable. But a lot of acid will injure these organs.

Possible complications include:

- **Esophagitis**. Esophagitis is inflammation in the lining of your esophagus. Chronic esophagitis can cause chronic pain and complications, like ulcers in your esophagus. After a long time, it can cause tissue changes like scarring or intestinal metaplasia, a precancerous condition.

- **Barrett's esophagus**. Barret's esophagus is the name for intestinal metaplasia of your esophagus. It means that the tissues lining your esophagus change to look like intestinal lining. This change happens after long exposure to acid and inflammation. It's a risk factor for esophageal cancer.
- **Esophageal stricture**. Your esophagus may also develop scar tissue to protect it from chronic inflammation and injury. Scar tissue can cause your esophagus to narrow. This is called stricture. Esophageal strictures can make it hard to swallow, which can make it hard to eat and drink.
- **Laryngopharyngeal reflux**. Some people with GERD also develop LPR, which is reflux that travels into your throat. Acid might sneak up into your throat while you sleep. It can cause swelling, hoarseness and vocal cord growths, and you can also aspirate acid particles into your airways.
- **Asthma.** Acid in your airways may aggravate existing asthma or cause asthma-like symptoms in people without any preexisting respiratory conditions. Tiny acid particles may irritate your bronchial tubes and cause them to contract, causing coughing and breathing difficulties.

Diagnosis and Tests

How is GERD diagnosed?

A gastroenterologist will evaluate your esophagus to diagnose GERD. Tests may include:

- **Esophagram**. An esophagram is a type of X-ray exam. It takes moving X-rays (fluoroscopy) of your esophagus while you swallow. For the test, you swallow a chalky liquid called barium.
- **Upper endoscopy**. An upper endoscopy involves looking inside your esophagus with a camera. The camera is on the end of a thin tube that passes through your mouth while you're asleep under light sedation.
- **Esophageal pH test**. This test measures the acid content inside your esophagus through a small wireless receiver. Your provider places the receiver in your esophagus during an endoscopy.
- **Esophageal manometry**. This test measures the muscle activity in your esophagus, using pressure sensors embedded in a nasogastric tube. It can confirm if your LES or other muscles aren't working right.

Management and Treatment

What is the medical treatment for GERD?

Some people find they can reduce acid reflux with lifestyle adjustments, like changing their eating habits, reducing alcohol and tobacco and losing weight. Healthcare providers encourage this approach.

But if you have chronic acid reflux or GERD, they also recommend medicines to reduce your stomach acid, so reflux is less damaging. These medicines are easily available and highly effective for GERD.

Medicine

Over-the-counter (OTC) medicines for acid reflux include:

- **Antacids**. Antacids (like Tums® and Rolaids®) neutralize your stomach acid so that when reflux happens, it isn't as corrosive to your esophagus. They work well for occasional acid reflux, but they can have side effects if you take them too often, so they aren't a good long-term solution.
- **Alginates**. Alginates are naturally occurring sugars derived from seaweed. They help block acid reflux by floating on top of the acid, creating a physical barrier between the acid and your esophagus. You can get alginates by themselves, and you can also get antacids with alginates.

Prescription medications for GERD include:

- **Histamine receptor antagonists (H2 blockers).** H2 blockers reduce stomach acid by blocking the chemical that tells your body to produce it (histamine). You can take them more frequently than antacids, but they don't always work long-term. Your body can adapt to their effects.

- **Proton pump inhibitors (PPIs).** PPIs are stronger acid blockers that also promote healing. Your provider may prescribe them as a first-line treatment if your GERD is relatively severe or you have signs of tissue damage in your esophagus. They're 90% effective in reducing acid reflux.
- **Baclofen.** Baclofen is a muscle relaxant, often prescribed to reduce muscle spasms. It can also help reduce the frequency of LES relaxation events, which reduces the frequency of acid reflux. Baclofen isn't a first-line treatment for acid reflux, but it may be a part of your treatment plan.

How do you get rid of acid reflux? Is there a cure?

While medicine can reduce the symptoms and the effects of GERD, it doesn't stop it from happening. A severe case of GERD may continue to cause complications despite medication, even if you can't feel it.

When this is the case, you need a more definitive treatment for acid reflux. This usually means a procedure to tighten your LES. These are minor, outpatient procedures, and they're very effective.

Surgery

Surgery for GERD includes:

- **Nissen fundoplication**. Fundoplication is the most common surgery for GERD. When possible, it's done as a laparoscopic surgery, which means smaller incisions and less recovery time. A surgeon wraps the top of your stomach around your lower esophagus and secures it with stitches to tighten the junction between them. This is also the procedure to fix a hiatal hernia.
- **LINX device.** A newer procedure implants a device called LINX during surgery. The LINX device is a ring of tiny magnets that help keep the junction between the stomach and esophagus closed.

Outlook / Prognosis

What can I expect if I have acid reflux?

So many things contribute to acid reflux that it can be hard to tell what's causing yours. Sometimes, simple lifestyle changes can make it go away. But some causes, like a hiatal hernia, may get worse.

If you have mild acid reflux, you can often manage it at home. If you have moderate to severe acid reflux, you might need a prescription to manage it. Medicine works for most people, but not all.

Whether your acid reflux bothers you a lot or a little, it's a good idea to discuss it with a healthcare provider. They can assess whether it's causing you complications that you may be unaware of.

If you have severe GERD or it causes complications that medicine can't help, you might need surgery. But surgery is usually minor and effective. It's worth treating GERD to prevent its complications.

Living With

What helps acid reflux at home?

To help manage acid reflux at home, try:

- **Eating smaller meals**. Larger meals expand your stomach and put pressure on your LES. Smaller meals digest faster and don't stimulate your stomach to produce so much acid.
- **Eating dinner earlier**. Gravity plays a role in keeping acid down, so it's a good idea to eat several hours before reclining in the living room or going to bed
- **Sleeping on your left side**. This positions your lower esophageal sphincter in an air pocket above your stomach contents. Lying on your back or your right side submerges the valve.
- **Reducing abdominal pressure**. Wear loose-fitting clothes or consider losing weight if you have overweight/obesity. This can help in the short term and the long term.

- **Quitting smoking and drinking**. Both tobacco and alcohol weaken your LES. They also affect your stomach, making it more acidic and slowing your digestion time.
- **Over-the-counter medications**. Antacids and alginates are good to have on hand, especially if you know you're eating a rich or acidic meal that might trigger more stomach acid.

What to do during an acid reflux attack?

If it's happening right now, try:

- **Standing up**. Gravity is on your side.
- **Taking a sip of water.** Don't drink a lot, but small sips may help wash the acid down.
- **Loosening your waistband**. Take off your belt or change your pants if it helps.
- **Taking an antacid**. If you don't have one, Pepto Bismol® might work.

When should I see my healthcare provider about acid reflux?

If you have acid reflux frequently, talk to a healthcare provider. It's important to find out how it's affecting your body. GERD isn't just an inconvenience — it can do real harm. It's also very treatable.

A note from Cleveland Clinic

Almost everyone gets acid reflux from time to time. Heartburn, indigestion and acid regurgitation can make for an unpleasant evening. You can take certain steps to reduce the frequency of these events.

If you have acid reflux often and it significantly affects your life, you might have GERD. It's worth talking to a healthcare provider about your symptoms. They can assess whether treatment might help.

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References

Cleveland Clinic's health articles are based on evidence-backed information and review by medical professionals to ensure accuracy, reliability and up-to-date clinical standards.

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